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Professor Shailesh Nayak
Editor-in-Chief
Journal of the Indian Society of Remote Sensing

Dear Professor Nayak,

We submit the Research Article, “Terrain-Conditioned Cross-Attention for Cross-Region Avalanche Debris Mapping with Sentinel-1 SAR,” for consideration in the Journal of the Indian Society of Remote Sensing.

The manuscript addresses a remote-sensing problem central to the journal’s scope: extracting hazard information from synthetic aperture radar under difficult terrain geometry. Sentinel-1 enables winter observation through cloud and darkness, but mountain foreshortening, layover, and shadow can resemble avalanche change. We therefore evaluate a terrain-conditioned change-segmentation model across seven events and four mountain systems, with geographically held-out tests in Tromsø and the Pamirs.

The contribution is both methodological and diagnostic. TopoRadar-Net couples bitemporal SAR features to local incidence angle, slope, aspect, elevation, and a disclosed fixed-reference aspect coordinate through multiscale cross-attention. More importantly, the paper tests this design against four baselines over three training seeds and reports negative evidence: the strongest baselines are statistically indistinguishable, some component removals improve pooled F1, and a slope prior suppresses genuine positives. A calibrated road-corridor proxy also exposes a complete miss on the only Tromsø corridor avalanche. These findings define where terrain conditioning helps and where it should not support operational claims.

The paper directly engages recent JISRS work on deep remote-sensing segmentation, SAR-specific learned representations, and disaster-debris mapping. Its distinction is the combination of cross-system SAR evaluation, spatial-block inference, multi-seed uncertainty, construct auditing, and explicit operational failure boundaries. The main text complies with the 4,000-word Research Article limit; a double-blind supplementary file contains detailed equations and complete secondary analyses.

This manuscript is original, has not been published, and is not under consideration elsewhere. Both authors approve its submission. The authors declare no competing interests and no external funding. Public data are cited by persistent DOI; the code and checkpoint archive are identified on the separate title page and can be provided confidentially during double-blind review.

We request standard double-blind peer review. We have not suggested or opposed any reviewer on the basis of expected opinion, nationality, or affiliation; any reviewer information entered in the submission system will follow the journal’s independence requirements.

Sincerely,

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